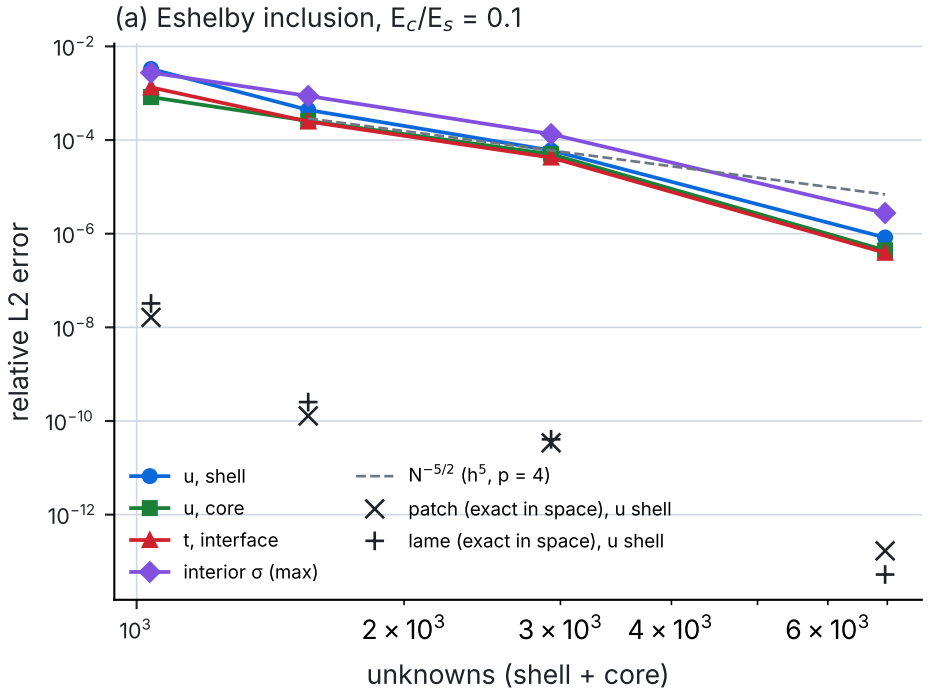
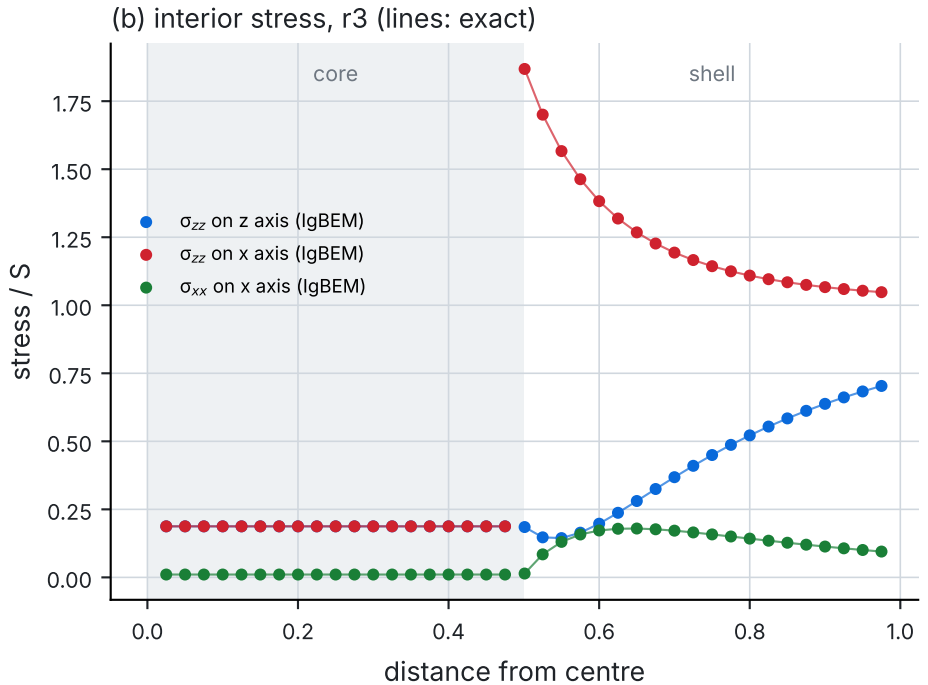
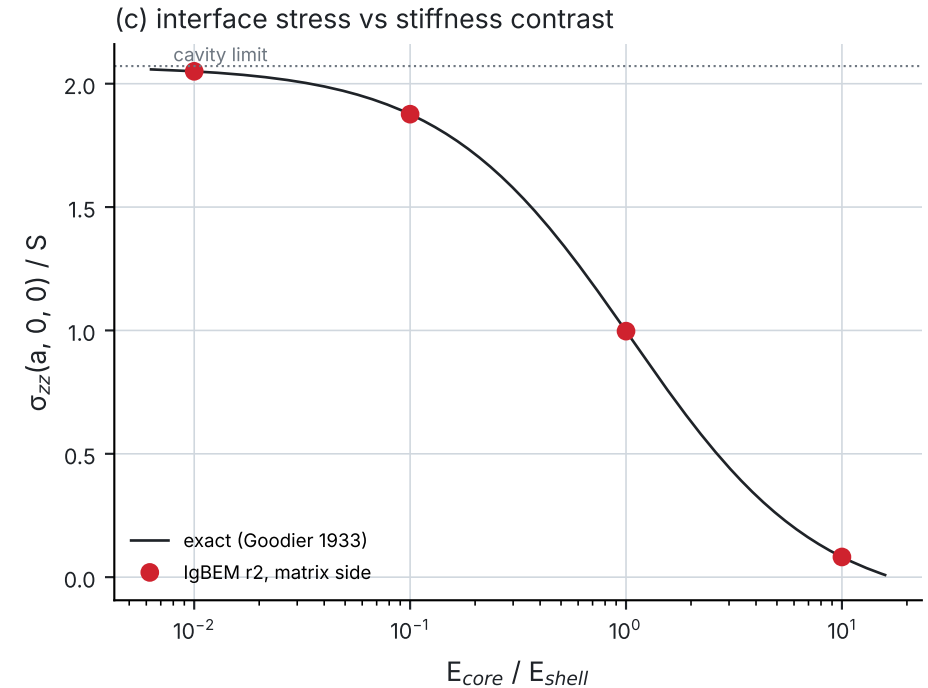
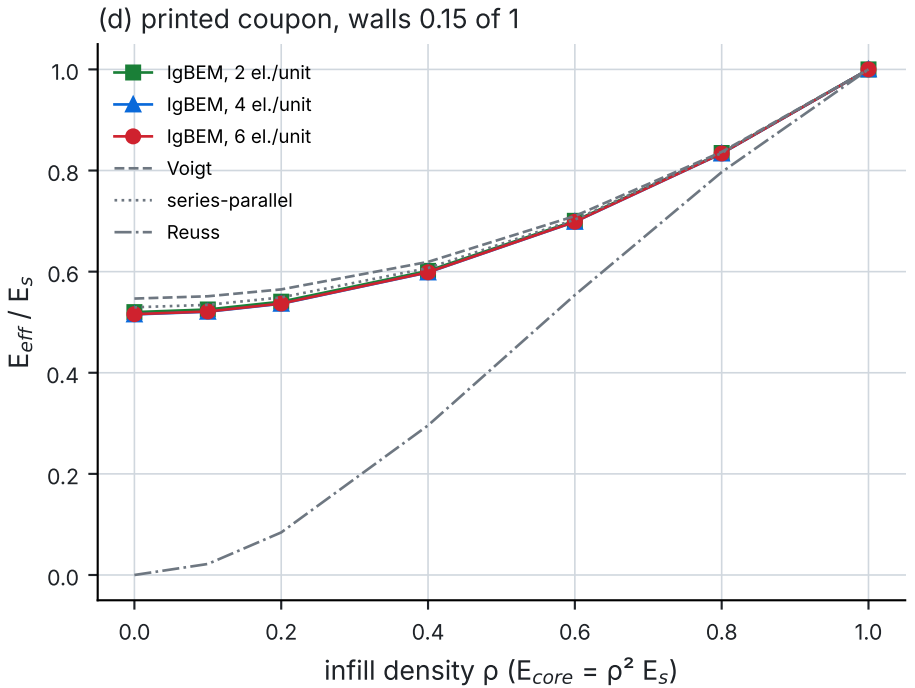

(a) Eshelby inclusion, E_c/E_s = 0.1
relative L2 error
unknowns (shell + core)
u, shell
u, core
t, interface
interior σ (max)
$N^{-5/2}$ (h^5, p = 4)
patch (exact in space), u shell
lame (exact in space), u shell


(b) interior stress, r3 (lines: exact)
core
shell
stress / S
distance from centre
σ_{zz} on z axis (IgBEM)
σ_{zz} on x axis (IgBEM)
σ_{xx} on x axis (IgBEM)


(c) interface stress vs stiffness contrast
cavity limit
$\sigma_{zz}(a, 0, 0)$ / S
E_{core} / E_{shell}
exact (Goodier 1933)
IgBEM r2, matrix side


(d) printed coupon, walls 0.15 of 1
E_{eff} / E_s
infill density ρ (E_{core} = ρ² E_s)
IgBEM, 2 el./unit
IgBEM, 4 el./unit
IgBEM, 6 el./unit
Voigt
series-parallel
Reuss